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To cite this article: Roman Farana, Pavel Brtva, Gareth Irwin & Karl M. Newell (2025) Neck Angle in the Handstand Changes the Pattern of Multi-Joint Variability, Journal of Motor Behavior, 57:6, 768-776, DOI: [10.1080/00222895.2025.2558711](https://doi.org/10.1080/00222895.2025.2558711)

To link to this article: <https://doi.org/10.1080/00222895.2025.2558711>



Published online: 22 Sep 2025.



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RESEARCH ARTICLE

Neck Angle in the Handstand Changes the Pattern of Multi-Joint Variability

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ABSTRACT. This study investigated the effects of neck angle on center of mass (CM) stability and joint angle variability in the handstand. Seven experienced female gymnasts performed handstands in extended, neutral, and flexed neck angles. Kinematic data were collected using a 3D motion capture system, and variability was assessed for wrist, elbow, shoulder, hip, and neck angles and, CM position. The findings showed that the extended neck angle posture exhibited the lowest CM variability, aligning with its prevalent use in practice and competition. Regression analyses revealed that joint angle variability was significantly correlated to CM motion, with the typical extended neck angle related to shoulder, wrist, and neck angle variability contributions (33%, 23%, and 21%, respectively). In contrast, the neutral neck angle was moderately correlated to hip variability (32%), while the flexed neck angle showed a dominant reliance on neck angle variability (63%). The results show the strong role of neck angle in the postural variability of the handstand that is mediated by joint angle variability, visual information and tonic reflex support.

Keywords: acrobatics, upright stance, balance, motor control

Introduction

The maintenance of balance in a standing posture is one of the fundamental motor skills that infants typically acquire through the sequence of prone progression in the first 18 months of life, and this skill continues to evolve across a range of environmental conditions and task demands throughout the lifespan (Haywood & Getchell, 2024; Thelen, 1995). Balance is essential for all postural activities, necessitating a coordinated organization of the central nervous system (CNS) to information obtained through proprioceptive, visual, and vestibular feedback systems (Kerwin & Trewartha, 2001). Postural control of limb segments is typically analyzed either through joint angle motions of the body as a kinematic chain or joint position in the external Euclidean space (Winter, 1995; Zatsiorsky, 2002).

The stability of a standing posture is contingent upon preserving the vertical projection of the center of mass CM within the area of the base of support, with the center of pressure (CP) reflecting the location on the support surface where collective neuromuscular input (reactive force) is used to control the motion of the CM (Winter, 1995). The maintenance of an upright stance has garnered significant research attention and has identified two primary strategies: the “ankle strategy” and the “hip strategy.” Nashner and McCollum (1985)

delineated the organization of an upright posture, highlighting a hierarchy of strategies that begins with the use of the ankle, followed by the addition of the hip, and, lastly, the knees. It has also been proposed that the goal of postural regulation in balance tasks is to stabilize the head in space (Roberts, 1973), but head motion has rarely been measured in experiments on postural control.

In addition to its role as a fundamental skill in gymnastics (George, 2010), the handstand (Figure 1) is a postural position that can contribute to our understanding of the underlying mechanisms in postural activities more generally (Asseman & Gahéry, 2005). The typical handstand is one where there is slight head extension, as reported by the FIG gymnastics Code of Points (FIG, 2025). Previous research has underscored that the limited stability region of the handstand necessitates a more constrained balance control strategy than that required for an upright stance (Clément & Rézette, 1985; Gautier et al., 2007). Maintaining a handstand posture has been interpreted as more complex because it involves at least four joints, wrists, elbows, shoulders, and hips, instead of 3, as for an upright stance (Blenkinsop et al., 2017). Gautier et al. (2007) reported that postural regulation in the handstand followed a system like that of an upright stance, with an interaction between 3 joints (wrist, shoulder, and hip) suggesting a common organization of human postures in spite of different muscular-skeletal organization.

The contrasting impact of these variables on predicting CM stability has been examined within the context of the anatomical or task-oriented organization of the structure of movement variability in joint angles and joint positions (Newell & Irwin, 2022; Pryhoda et al., 2022). An inverted stance (handstand) is less frequently practiced and inherently also less stable due to a smaller area of the base of support, higher CM position from the base of support (Wyatt et al., 2021), and reduced capacity to generate torques about the wrist compared to the ankle. Blenkinsop et al. (2017) found that during various perturbed conditions (closed eyes and platform vibration in forward and backward directions) and unperturbed conditions (open eyes) in the handstand, the most prevalent

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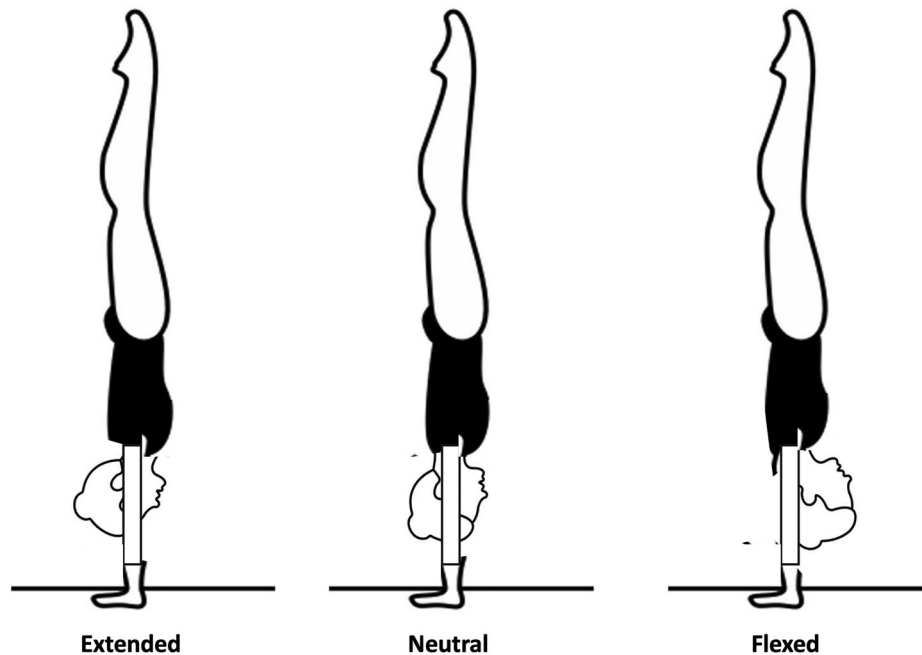


FIGURE 1. Schematic illustration of the different neck angles performed by the gymnasts.

control strategy was a wrist strategy, employed for over 75% of the time during balance. Wyatt et al. (2021) found that skilled performers also exploited mechanical advantages in the mediolateral direction, such as a wider base of support, compared to less skilled gymnasts.

Movement variability has been postulated to facilitate the pick-up of information of the perceptual-motor workspace and to be directly linked to improved task performance (Newell et al., 1989; Riccio, 1993). The role of vision (Gautier et al., 2007) and neck angle (defined as head to trunk angle) (Asseman & Gahéry, 2005) in the stability of handstand skills has been investigated. Asseman and Gahéry (2005) showed that performance and postural stability were significantly better in the typical dorsiflexion neck angle than in the aligned and ventroflexion positions under both vision and non-vision conditions. It was proposed that the extended neck angle, unlike the neutral or flexed angle, affords the visual focus to be coordinated to the projected surface location of the CM.

The neck angle could also influence postural stability in the handstand through gradations of symmetrical tonic reflex support of the relevant neural pathways. Indeed, it is a long-standing hypothesis that balance in postural control is modulated by reflexes (Asseman & Gahéry, 2005; Fukuda, 1984; Hellebrandt et al., 1956, 1962; Roberts, 1967; Zehr & Stein, 1999), but relevant experimental examination in humans has been very

limited. A reflex interpretation provides additional support to the positive influence of visual information on the balance of an extended neck angle in the handstand posture.

Previous research has emphasized that the limited stability region of the handstand necessitates a more constrained balance control strategy than that for an upright stance (Clément & Rezette, 1985; Gautier et al., 2007). Pryhoda et al. (2022) found less variability in upright standing in terms of both amount and time-dependent structure compared to the handstand. They suggested that the patterns of postural variability indicate a task-specific nature in joint motions rather than an anatomical pattern of organization.

The role of neck angle in human movement and postural control has not been extensively studied in an experimental framework, even in gymnastic tasks that are assumed by gymnasts and coaches to be influenced by it. Therefore, the primary aim of this study was to determine the behavioral effect of the manipulation of neck angle on postural joint variability in the handstand. The movement variability of the wrist, elbow, shoulder, hip, and neck joint angles, and CM position was contrasted in handstands with the neck angle in one of three postural forms (extended, neutral, flexed). We hypothesized that the extended neck angle would be associated with lower center of mass variability compared to neutral and flexed head positions, reflecting more stable balance.

Methods

Participants

Seven female gymnasts, experienced in balancing in a handstand, were recruited for this study (mean age 21.0 ± 1.9 yrs, mean height 162.0 ± 4.4 cm, and mean mass 55.8 ± 5.1 kg). All gymnasts had more than 10 yrs of experience in gymnastics training and competitive gymnastics at the national level. They were able to maintain balance in the handstand position for at least 30 s (Blenkinsop et al., 2017). All gymnasts were injury-free at the time of testing, and informed consent was obtained under the guidelines of the Institute's Ethics and Research Committee.

Procedures and Data Collection

The gymnasts completed their self-selected warm-up and practiced handstands with different neck angles. After the warm-up and practice, all gymnasts performed three trials of handstands with slight neck extension, neutral, and maximum neck flexed neck angles (Figure 1). To ensure a valid handstand performance the mean neck angle was used to define the neck angle. The neck angle was calculated and showed low levels of within and between gymnast differences for extended = $191.96 \pm 1.10^\circ$, neutral = $178.44 \pm 1.21^\circ$, and flexed = $164.40 \pm 1.13^\circ$ neck angles. These differences indicate that the gymnasts maintained the required neck angle throughout the handstand movement task, as reflected by the consistent alignment across the three conditions. Throughout this study the neck angle was defined as the sagittal angle between the center of the of the head and trunk segment.

Based on the information obtained, all gymnasts primarily used the typical extended handstand neck angle in their training and competitions, while the other two positions (neutral and flexed) were new to them as movement tasks. All trials were performed in a random order and were separated by a 1-min rest period. Data were recorded for each gymnast executing a series of handstand balances lasting 5 s, and these data were used for further analysis (Kerwin & Trewatha, 2001). Gymnasts were instructed to maintain a static base of support and attempt to remain in or return to the standard starting position of fully extended arms, trunk, and legs with feet together for handstand trials. This procedure was previously used in the study by Blenkinsop et al. (2017).

Kinematic data were recorded using 16 Qualisys cameras (Qualisys Oqus, Sweden) sampling at 240 Hz. In total, 36 active markers and eight clusters with four markers were bilaterally fitted to each gymnast on the lower extremities, pelvis, lumbar spine, torso, upper extremities, and head (Farana et al., 2023). Participants wore shorts and a cropped top to allow for maximal marker placement directly on the skin. All handstand

trials were evaluated by a national-level coach to ensure their correct execution in accordance with FIG rules, with the handstand position considered reached when all body parts are aligned vertically (FIG, 2025).

Data Processing and Analysis

Kinematic data were calculated in Visual 3D software (v6, C-motion, Inc., Rockville, MD) for the inverted stance durations of three trials for each gymnast. The whole-body CM was calculated as a weighted average of 16 segments (feet, shanks, thighs, pelvis, lumbar spine, torso, hands, forearms and upper arms, and head) using individual Visual 3D models, following Wyatt et al. (2021). Based on a previous study by Kerwin and Trewatha (2001), each gymnast was represented in the sagittal plane by averaging the left and right joint center coordinates and comprising a hand, arm, trunk, head, and leg segment, with five corresponding joints: wrist, elbow, shoulder, hip, and neck. Joint angles were used to describe the position of the CM during the handstand, as they provide insight into how segmental changes in body posture influence the overall CM position and stability, despite joint angles and CM representing different frames of reference. By analyzing joint angles, we can understand how local changes in body posture impact the global position of the CM, despite these different reference systems.

The standard deviation (SD) was calculated from the time series during the phase in which the gymnast held a handstand. The data were recorded continuously, allowing for the capture of fluctuations in the gymnast's body position throughout the entire duration of the handstand. The standard deviation calculation thus reflects the movement variability within this time series (time-series variability), where each measurement point was analyzed individually (point-by-point). This quantified the level of stability or dynamic change during the performance.

Statistical Analysis

Gymnast-specific skill variability was defined as the mean standard deviation in each kinematic measure for each handstand trial performed by the respective gymnast. The group mean standard deviation was determined ($n=7$ gymnasts) for each respective handstand technique. Mean standard deviations for joint angles and CM positions across the three trials were calculated, and then group means were determined and subsequently used for statistical analysis. Variability was calculated and expressed as the joint flexion/extension standard deviation across trials.

Using SPSS software (IBM SPSS Statistics 25, SPSS Inc., Chicago, IL), the statistical differences of wrist, elbow, shoulder, hip, neck, and CM variability between the typical extended, neutral, and flexed neck angle were

analyzed using One-way ANOVA, followed by Holm corrections. Effect sizes (ES) were calculated for the ANOVA *via* partial η^2 and presented as <0.01 trivial; 0.01–0.06 small; 0.06–0.14 moderate, and >0.14 large (Cohen, 1992). In addition, Effect sizes were interpreted using Cohen's *d* (Cohen, 1992) with thresholds for trivial (0–0.2), small (0.2–0.6), moderate (0.6–1.2), large (1.2–2.0), and very large (>2.0) effects. Additionally, the strength of correlations was classified using Pearson's *r*, with ranges indicating trivial (0.0–0.1), small (0.1–0.3), moderate (0.3–0.5), large (0.5–0.7), very large (0.7–0.9), and perfect (1.0) relationships. Linear multiple regressions were conducted to explore the relationship between CM and joint variability. CM position variability was the dependent variable, with joint variability as an independent variable. This analysis determined which joint variability was prominently related to CM position variability. To express the contribution of each predictor to the model, we calculated the percentage of variance explained by each variable. This was done by computing the ratio of the standardized coefficient squared for each predictor to the total explained variance, expressed as a percentage. For all tests, statistical significance was set at $p < .05$.

Results

The individual gymnast standard deviation for the joint angles' variability and CM position variability measures are presented in Table 1 and reveal differences in joint variability and CM position across joint positions.

The one-way ANOVA revealed a significant main effect for CM variability (SD) among the three neck angles ($p = .02$, $\eta^2 = .35$), indicating that the handstand with the extended neck angle had the lowest CM variability (Figure 2). Subsequent pairwise comparisons

showed significant disparities with substantial effect sizes (ES) between the extended and neutral neck angles ($p = .04$, ES = 1.3) and between the extended and flexed neck angles ($p = .03$, ES = 1.5), respectively. No significant differences in neck joint variability were observed, but a large effect size was found between flexed and extended neck angles (ES = 1.0) (Figure 2).

A significant main effect was observed for shoulder joint angle variability ($p < .001$, $\eta^2 = .94$). Subsequent pairwise comparisons revealed significant differences, with substantial effect sizes (ES), between the extended and neutral neck angles ($p < .001$, ES = 7.0), and between the extended and flexed neck angles ($p < .001$, ES = 3.8) (Figure 3). Likewise, a significant main effect was obtained for hip joint variability ($p = .01$, $\eta^2 = 0.39$). Significant differences and notable effect sizes were found in the hip joint between the neutral and flexed neck angles ($p = .004$, ES = 2.0). While no statistically significant differences were observed, there were substantial effect sizes between the extended and neutral neck angles (ES = 1.0) and between the extended and flexed neck angles (ES = 1.6) (Figure 3). In contrast, no significant differences in wrist and elbow joint variability were identified among the various neck angles (Figure 4).

In addition, Table 2 shows a correlation between the dependent variables for different neck angles. In the extended neck angle, several moderate to large correlations were observed. The correlation between CM position and wrist joint variability was moderate ($r = 0.48$), and the correlation between CM position and shoulder joint variability was also moderate ($r = 0.36$). The relationship between hip joint and shoulder joint variability showed a very large positive correlation ($r = 0.88$), indicating a strong relationship between these variables.

TABLE 1. Gymnast-specific and group standard deviation for the wrist, elbow, shoulder, and hip joint measures.

	Wrist (°)			Elbow (°)			Shoulder (°)			Hip (°)			Neck (°)			CM (mm)		
	Ext	Neu	Fle	Ext	Neu	Fle	Ext	Neu	Fle	Ext	Neu	Fle	Ext	Neu	Fle	xt	Neu	Fle
GYM1	2.7	4.1	1.7	1.2	1.8	1.1	1.2	1.8	1.9	1.9	2.0	1.6	1.3	1.4	0.5	14.2	17.7	15.7
GYM2	1.2	1.2	2.3	1.9	1.3	2.6	1.2	1.7	1.9	1.7	1.8	1.4	1.0	1.0	0.7	9.9	12.2	18.8
GYM3	2.5	3.0	2.6	2.0	2.0	1.8	1.0	1.6	1.7	1.4	1.7	1.3	0.5	1.6	2.1	9.9	15.0	17.4
GYM4	1.8	2.9	2.4	0.6	1.4	0.7	1.0	1.7	1.4	1.4	1.8	1.2	0.5	0.5	0.8	7.5	11.2	7.6
GYM5	0.8	1.0	1.1	2.2	0.7	1.2	1.0	1.7	1.5	1.3	1.5	1.1	0.8	1.1	1.0	7.4	14.5	17.4
GYM6	1.6	2.1	2.7	1.7	2.3	1.1	1.0	1.8	1.4	1.3	1.5	1.2	0.6	1.4	4.0	13.2	22.7	23.6
GYM7	0.7	0.8	1.3	1.1	2.1	2.8	0.9	1.7	1.6	1.3	1.4	1.3	0.7	1.2	1.5	11.0	16.9	13.9
GROUP	1.6	2.2	2.0	1.5	1.7	1.6	1.0	1.7	1.6	1.5	1.7	1.3	0.8	1.2	1.5	10.5	15.7	16.3
SD	0.7	1.1	0.6	0.5	0.5	0.7	0.1	0.1	0.2	0.2	0.2	0.2	0.3	0.3	1.1	2.4	3.5	4.5

Notes: Ext, extended ; Neu, neutral; Fle, flexed; H, hip; S, shoulder; E, elbow; W, wrist; N, neck.

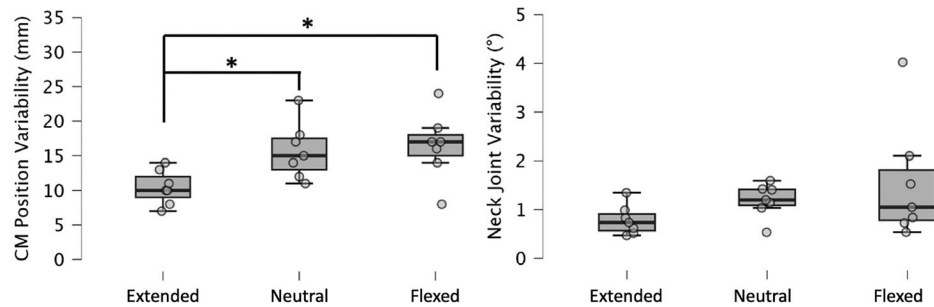


FIGURE 2. Center of mass position and neck joint variability for different neck angles: 1 (extended), 2 (neutral), and 3 (flexed).

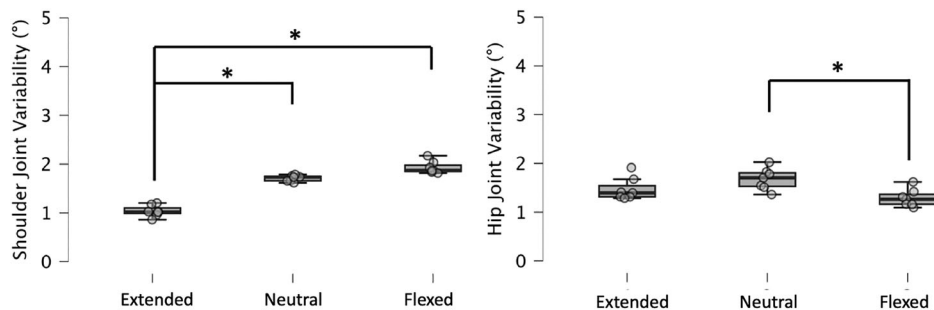


FIGURE 3. Shoulder and hip joint variability for different neck angles: 1 (extended), 2 (neutral), and 3 (flexed).

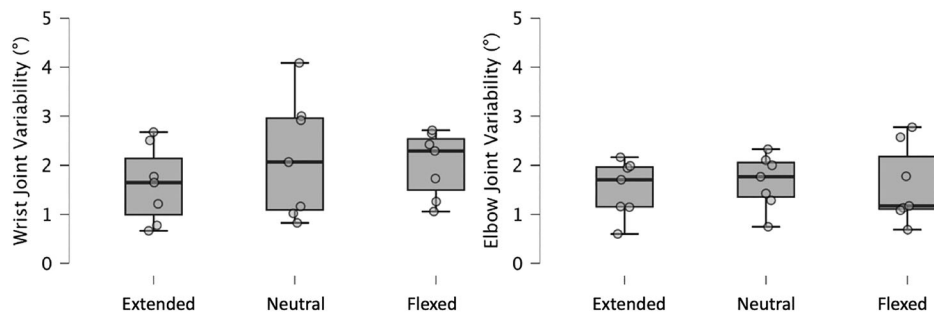


FIGURE 4. Wrist and elbow joint variability for different neck angles: 1 (extended), 2 (neutral), and 3 (flexed).

In the neutral neck angle, the correlation between CM position variability and shoulder joint variability was large ($r = 0.75$), while the relationship between CM position variability and elbow joint variability showed a large correlation ($r = 0.71$). Additionally, the correlation between wrist angle variability and hip joint variability was large ($r = 0.73$).

In the flexed posture, the correlation between hip joint variability and shoulder joint variability remained large ($r = 0.66$), while several negative correlations emerged, such as the relationship between shoulder joint variability and neck joint variability ($r = -0.52$), indicating a large inverse relationship.

Table 3 presents the outcomes of multiple regression analyses performed to assess the influence of neck angle

TABLE 2. Correlation between measured variables for different neck angles.

Variables	Extended	Neutral	Flexed
CM-W	0.48*	0.12	0.22
CM-E	-0.16	0.71***	0.15
CM-H	0.48*	-0.31	0.02
CM-S	0.36	0.75***	0.26
CM-N	0.46*	0.67**	0.65**
W-E	-0.17	0.27	-0.22
W-H	0.55**	0.73***	-0.06
W-S	0.50*	-0.11	-0.12
W-N	0.14	0.17	0.48*
E-H	-0.14	-0.22	0.30
E-S	0.28	0.19	0.41*
E-N	0.01	0.50*	-0.12
H-S	0.88***	-0.27	0.66**
H-N	0.84***	-0.09	-0.44
S-N	0.72***	0.25	-0.52**

Notes: CM, center of mass; H, hip; S, shoulder; E, elbow; W, wrist; N, neck.

*Moderate relationship.

**Large relationship.

***Very large relationship.

TABLE 3. Multiple regressions of CM for different neck angles.

Head position	R ² in final model	Joint %predictor
Extended (1)	0.88*	S(33%), W(23%), N(21%), H(12%), E(11%)
Neutral (2)	0.96*	H(32%), W(23%), N(20%), E(13%), S(12%)
Flexed (3)	0.91*	N(63%), W(17%), S(13%), E(4%), H(3%)

Notes: H, hip; S, shoulder; E, elbow; W, wrist; N, neck.

*= $p < .05$.

on the variability of joint angles to identify predictor variables for the center of mass (CM) motion in three different neck angles.

The adjusted R^2 values in the final models for each neck angle provide insights into the goodness of fit for the regression models. For the extended neck angle, the adjusted R-squared value was 0.88, indicating that the model, incorporating wrist (W), elbow (E), shoulder (S), hip (H), and neck (N) angles, correlates to 88% of the variance in CM motion. The joint predictor, in terms of weighting contribution, was S (33%), W (23%), N (21%), H (12%), W (22%), and E (11%). These findings

emphasize the significant role of shoulder, wrist, and neck angles in explaining CM motion for this neck angle.

The model displayed an even higher adjusted R^2 value of 0.96 for the neutral neck angle. Predictor variables were H (32%), W (23%), N (20%), E (13%), and S (12%). The substantial R^2 value and the dominant contributions of hip, wrist, and neck angles underscore the strong predictive power of this model for the neutral neck angle. Lastly, for the flexed neck angle, the adjusted R^2 value was 0.91, with predictors of importance being N (63%), W (17%), S (13%), E (4%), and H (3%), which showed that neck angle played the strongest role in relating to CM variability for this neck angle.

Discussion

The primary objective of mastering the handstand skill is to maintain balance, a task that becomes particularly challenging when the neck angle is altered (Asseman & Gahéry, 2005; Gautier et al., 2007). The current study investigated how the CM variability in the handstand was influenced by different neck angles (extension, neutral, flexion) and determined the contributions of the wrist, elbow, neck, shoulder, and hip joint angle variabilities to the variability of the motion of the whole-body CM.

The results showed that the handstand with the typical extended neck angle had lower CM variability than that for a neutral or flexed neck position (Figure 2). This finding is consistent with Asseman and Gahéry (2005), who observed that postural stability was improved in the extended dorsiflexion positions compared with neutral (aligned) and ventroflexion (flexion) positions. The typical extended neck angle is the most common technique during practice and competition and this head technique follows coaching guidance (Broomfield, 2011; Mitchell et al., 2002).

The manipulation of neck angle changes the focus of vision and hence the pick-up of information in maintaining balance in handstand postures (Gautier et al., 2007). In parallel, it also shows the effect of neck angle and the associated neck reflex to the stability of balance. Asseman and Gahéry (2005) proposed that in the typical dorsiflexed neck angle neck reflexes generate a tonic activity of the extensor limb muscles (Fukuda, 1984). This facilitated muscular activity helps the arms to support the body weight and makes it easier to maintain the vertical position of the body in the handstand.

The findings from the regression analysis showed the relationship between neck angle joint variability and its implications for postural control during an inverted stance. Specifically, the regression models demonstrated the contribution of joint angle variability to the variability of the whole-body CM motion under different neck

angles. In the extended neck angle, approximately 88% of the variability in CM motion was related to the joint angle variability of the shoulder, wrist, neck, hip, and elbow. These findings support the view that, in the extended neck angle, shoulder angles play the most substantial role (33%) in relating to CM motion variability, followed by wrist angles (23%), neck angles (21%), hip angles (12%), and elbow angles (11%) (Table 3). These findings concur with Gautier et al. (2007), who proposed that gymnasts implemented a multiple joint strategy.

The participants in this study were skilled and will most likely have mastered the redundant degrees of freedom and may be in a state of “freeing” the degrees of freedom involved in the movement task (Bernstein, 1967). This is counter to the finding of Blenkinsop et al. (2017), who noted the prevalence of a wrist strategy during handstand control under different conditions, suggesting its significance for balance maintenance. Also, this implies that for the extended neck angle, maintaining postural control primarily relies on an increase in dimensionality provided by the multiple joint system (Newell & Vaillancourt, 2001). It also aligns with the concept of proximal-to-distal control, where the control of movement initiates at proximal joints and progresses toward distal ones (Newell & Irwin, 2022). Contrary, when the neck angle is in a neutral position, the hip joint dominates as the most influential predictor, followed by the wrist and neck joint, suggesting an adaptation to the specific demands of this neck angle.

This shift in the order of joint importance of the kinematic chain aligns with the idea that dimensionality increases in response to the neutral neck angle, potentially optimizing stability and balance. This hip strategy was observed in the previous study by Blenkinsop et al. (2017), as less of the prevalence strategy during handstand control under different conditions. Conversely, the extended neck angle displays a different pattern, with the neck playing a prominent role. This adaptation may be indicative of a need to decrease the dimensionality related to accommodate the specific requirements of the flexed neck angle, showing the adaptability of the motor control system to task-specific demands. Moreover, in all three neck angles, the wrist joint plays an important role as the second-highest predictor, indicating that movement control initiates at the distal wrist joint in response to the unique demands posed by the different neck angles (Slobounov & Newell, 1996).

These findings lend support to the notion that the dimensionality of task-relevant intrinsic dynamics can either increase or decrease based on the constraints imposed by neck angle. When facing new task demands, the motor system appears to adapt by altering the relative importance of joint angles and their collective dimensionality (Newell & Vaillancourt, 2001). This insight enhances our understanding of motor control in different neck angles and the dynamic adjustments made to

maintain stability and meet the specific demands of each posture (Asseman & Gahéry, 2005). These findings underline the significance of neck angle in influencing the relative importance of joint angle variability in explaining CM motion variability (Wyatt et al., 2021). The shifting order of joint inclusion and their respective percentage contributions emphasize the dynamic nature of this relationship, highlighting the intricate interaction between neck angle and joint angle variability in the context of maintaining balance and stability during inverted stances. This interpretation also relates to the broader view of motor coordination as emerging within a low-dimensional movement manifold (Daffertshofer et al., 2004). Dimensionality reduction methods such as principal component analysis (PCA) have been widely used to extract dominant modes of coordination from high-dimensional motor data (Deluzio et al., 1997; Donà et al., 2009; Dotov et al., 2025), while the residual variance often reflects meaningful aspects of intrinsic variability rather than mere noise. From this perspective, the observed increases in dimensionality under neutral neck angles may indicate a broadening of the coordination manifold to support flexible balance strategies, whereas decreases in dimensionality under flexed or extended neck angles suggest a tighter coupling of joint actions to stabilize posture under more constrained conditions.

These findings have practical implications for activities involving inverted stances, emphasizing how neck angle influences joint contributions to postural control. For gymnasts and coaches, understanding these biomechanical adjustments is crucial for effective training and performance. This adaptability underscores the dynamic nature of motor learning and the intricate interplay between neck angle, joint angle variability, and postural control in gymnastics. The predominance of the extended neck angle among gymnasts aligns with its stability in terms of CM motion variability, mainly influenced by hip angles. This underscores the practical relevance of training regimens and coaches' focus on neck angles.

Moreover, gymnasts' adaptation to less familiar neck angles (neutral and flexed) demonstrates their adaptability in handling novel challenges, emphasizing the importance of integrating neck angle considerations into handstand training programs for enhanced postural stability control. These findings suggest that the motor system adapts by altering the relative contribution of joint angles based on the neck angle to minimize CM variability. This enhances our understanding of motor control across different neck angles, reflecting the intricate interplay between proximal-to-distal and distal-to-proximal control according to task demands (Newell & Irwin, 2022).

The variability data reveal distinct strategies individual gymnasts employ to maintain balance under different neck angles. Gymnast 1, for example, exhibited a clear reliance on wrist adjustments, as indicated by higher

wrist variability in the neutral position (4.1°) compared to extended (2.7°) and flexed (1.7°). This suggests a distal control strategy to compensate for reduced head and neck stability. In contrast, Gymnast 5 displayed a highly stable wrist strategy with consistently low variability across all neck angles, highlighting a potential preference for proximal control mechanisms through the shoulder and hip. Gymnast 6, who demonstrated the highest neck variability in the flexed position (4.0°), appears to rely heavily on head and neck adaptations when faced with challenging postural demands, potentially indicating less reliance on distal joints like the wrist. On the other hand, Gymnast 3 exhibited minimal hip variability in the flexed position (1.3°), reflecting strong proximal stability that likely reduced reliance on distal adjustments. These individualized approaches illustrate how gymnasts tailor their motor strategies to their strengths and to the specific demands of each neck angle. By understanding these strategies, coaches can develop tailored training plans to address specific needs. Such targeted interventions could optimize performance by aligning training with each gymnast's unique motor strategy.

The observed differences between neck angles likely reflect the combined influence of biomechanical alignment, sensory input, and motor learning. Changes in neck angle modify segmental geometry, altering mechanical demands for balance control (Blenkinsop et al., 2017). Simultaneously, shifts in gaze direction may affect visual feedback critical for spatial orientation and postural stability (Gautier et al., 2007). Finally, the extended head position, being more typical in competitive practice, may reflect a learned motor solution optimized through task-specific experience (Newell, 1991). Future studies should control visual conditions and provide equal familiarization across postures to better isolate these interacting factors.

Limitations

This study has several limitations that should be acknowledged. First, while our findings suggest that reflexive, proprioceptive, and biomechanical mechanisms contribute to the observed effects of head position on postural stability, the potential role of altered visual input cannot be excluded. Changes in head orientation necessarily affect gaze direction and visual field, which may influence the perception of body orientation and support surface. Future studies should control for visual input using gaze fixation, visual occlusion, or eye-tracking protocols to better isolate the specific sensory contributions to postural regulation. Second, the relatively small sample size of elite gymnasts, primarily determined by feasibility constraints, is consistent with previous research in this population (Asseman & Gahéry, 2005; Blenkinsop et al., 2017; Gautier et al., 2007), but limits statistical power and generalizability.

Larger samples would enable a more detailed analysis of individual differences, including responder classification or cluster analysis to identify distinct coordination strategies across gymnasts. Finally, while we analyzed joint-specific variability and contributions to CM control, we did not directly examine inter-joint coordination patterns. Future work incorporating dimensionality reduction methods such as principal component analysis (PCA) or uncontrolled manifold (UCM) analysis could further clarify how joint variability is organized to stabilize the center of mass during inverted balance tasks.

Conclusions

In summary, this investigation shows the pivotal role played by neck angle in channeling the variability of joint angles of the kinematic chain and, consequently, influencing CM motion during handstands. The findings reveal the significant impact of neck angle on CM variability, resulting in distinct patterns of joint angle contributions to postural control. The typical slightly extended neck angle, a prevalent choice in gymnastic practices, emerges as a stable option, with shoulder angles prominently correlated with CM motion. In contrast, the neutral neck angle prioritizes hip stability, presenting a shift in the hierarchy of joint importance. In the flexed neck angle, neck angles take precedence, reflecting the heightened reliance on changes in neck angle for postural control.

The dynamic order of joint inclusion and their respective percentage contributions accentuates the intricate interaction between neck angle and joint angles in the context of maintaining balance and stability during handstands. This study highlights the individualized motor strategies gymnasts use to maintain balance during handstands under different neck angles. While general trends, such as higher wrist variability in the neutral position and increased neck variability in the flexed position, were evident, each gymnast demonstrated unique strategies. Some relied on distal joints for control, while others prioritized proximal stability, reflecting flexibility between distal-to-proximal and proximal-to-distal control sequences.

Disclosure Statement

No potential conflict of interest was reported by the author(s).

Funding

This work was supported by the European Union project LERCO (CZ.10.03.01/00/22_003/0000003).

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Received February 14, 2025

Revised September 1, 2025

Accepted September 2, 2025